

Esteban Sitek

Senior Infrastructure Engineer, PostgreSQL

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PROFESSIONAL SUMMARY

Senior Infrastructure Engineer with **12+** years across fintech, healthcare, and e-commerce, building resilient cloud platforms and database systems at scale. Expert in PostgreSQL production operations, Kubernetes orchestration, and Infrastructure as Code. Experienced designing highly available multi-region AWS infrastructure, automating operational workflows with Go-based controllers, and mentoring teams on database reliability best practices. Ready to own Laravel Cloud's managed PostgreSQL offering end-to-end.

SKILLS

- Languages:** Go, Python, JavaScript, TypeScript, Bash, SQL, HTML, CSS
- Infrastructure & DevOps:** Kubernetes, AWS, Terraform, Docker, PostgreSQL, Argo CD, Helm, ECS, EC2, S3, RDS, CloudFormation, VPC, IAM, Route 53, CloudFront, CloudWatch
- Database:** PostgreSQL, Replication, Backup & Recovery, High Availability, Performance Tuning, Schema Migrations, Monitoring, Indexing
- Operators & Controllers:** Kubebuilder, Controller-runtime, Operator SDK, CloudNativePG, Crunchy Postgres for Kubernetes, Custom Controllers
- Observability & Monitoring:** Prometheus, Grafana, OpenTelemetry, ELK Stack, Datadog, Logging, Alerting
- Backend:** Node.js, Express, NestJS, FastAPI, Django, Laravel, gRPC
- Frontend:** React, Next.js, Vue, Redux, Tailwind CSS, Material UI
- Testing:** Jest, Pytest, Go Testing, Integration Testing, Load Testing
- CI/CD & GitOps:** GitHub Actions, GitLab CI, Argo CD, Flux, Jenkins, CircleCI
- Cloud & DevOps:** AWS, Kubernetes, Docker, Terraform, Helm, ECS, EC2, S3, RDS, CloudFormation, VPC, IAM, Route 53, CloudFront, CloudWatch, Linux, Networking
- Others:** RESTful API, GraphQL, Git, GitHub, Jira, Confluence, DNS Management, Load Balancing, Caching

EXPERIENCE

Publicis Sapient

Senior Software Engineer

Jun 2024–Present

- Spearheaded the design and deployment of a multi-region **AWS** infrastructure on **Kubernetes** clusters, implementing **Terraform** templates to manage infrastructure across development, staging, and production environments while ensuring compliance and security policies.
- Owned the PostgreSQL database operations for a high-throughput financial services platform, implementing replication, automated backups, and point-in-time recovery mechanisms that reduced recovery time objective (RTO) from 8 hours to under 30 minutes.

- Built a **Go**-based **Kubernetes** operator using **Kubebuilder** and **controller-runtime** to automate stateful workload management, eliminating 12+ hours of monthly manual operational overhead and increasing cluster reliability by 99.9%.
- Implemented comprehensive monitoring and alerting using **Prometheus** and **Grafana**, instrumenting PostgreSQL metrics, Kubernetes node health, and AWS resource utilization across 15+ microservices, enabling real-time incident detection.
- Collaborated with platform engineering teams to establish Infrastructure as Code standards, enforcing **Terraform** policies through automated validation pipelines that prevented configuration drift and reduced production incidents by 40%.
- Designed and implemented a Disaster Recovery strategy across multiple AWS regions, including automated failover mechanisms for PostgreSQL using streaming replication and RTO/RPO targets aligned with SLA requirements.
- Mentored junior infrastructure engineers on Kubernetes best practices, high availability design patterns, and PostgreSQL performance tuning techniques through peer code reviews and knowledge-sharing documentation.
- Troubleshooted and resolved critical production issues involving Kubernetes node failures, PostgreSQL connection pool exhaustion, and AWS networking bottlenecks, maintaining 99.95% uptime during periods of 2M+ daily transactions.
- Implemented **GitOps** workflows using **Argo CD** for declarative infrastructure deployments, enabling the team to deploy configuration changes with full auditability and automatic rollback capabilities.
- Hardened database security posture by implementing row-level security policies in PostgreSQL, encrypting data at rest using AWS KMS, and establishing least-privilege access controls through IAM roles and **Kubernetes** RBAC.
- Automated PostgreSQL major version upgrades across production clusters, developing custom **Go** automation scripts that coordinated zero-downtime migrations affecting 500+ application instances.

CareSimple

Oct 2022–May 2024

Senior Full Stack Engineer

- Led the migration of the patient data platform from a monolithic **PostgreSQL** instance to a sharded architecture on **AWS**, implementing **Terraform** IaC to manage multi-account infrastructure and reducing query latency by 65% for patient record retrieval.
- Developed and maintained a **Go**-based data synchronization service that orchestrated ETL pipelines across healthcare data warehouses, ensuring HIPAA compliance and processing 50M+ patient records daily with 99.99% delivery accuracy.
- Owned the design and operation of a **Kubernetes** cluster hosting patient monitoring services, implementing network policies, persistent volume strategies, and resource quotas that enabled safe multi-tenant workload isolation across healthcare providers.
- Implemented high availability for PostgreSQL using streaming replication and automated failover with custom **Go** controllers, reducing unplanned downtime incidents from 4–5 per quarter to zero over the next 12 months.

- Built comprehensive observability infrastructure using **Prometheus**, **Grafana**, and **OpenTelemetry**, creating dashboards and alerts for critical paths like patient notification delivery and result reporting, cutting mean time to detection (MTTD) by 70%.
- Collaborated with backend teams using **Node.js** and **FastAPI** to optimize database query patterns, reducing N+1 queries and implementing caching strategies that decreased PostgreSQL CPU utilization by 35% during peak hours.
- Established disaster recovery procedures for sensitive healthcare data, implementing automated PostgreSQL backups to **S3** with encryption, cross-region replication, and quarterly recovery drills that met RTO/RPO SLA targets.
- Mentored full stack engineers on infrastructure best practices, containerization strategies, and security fundamentals, conducting 10+ workshops on **Kubernetes** networking and PostgreSQL troubleshooting.
- Deployed and managed CI/CD pipelines using **GitHub Actions** and **Argo CD**, enabling the team to ship database schema migrations and application updates with confidence and automated rollback on health check failures.

Shopee

Jul 2018–Sep 2022

Full Stack Engineer

- Played a key role in building and operating a geographically distributed **PostgreSQL** infrastructure serving the Southeast Asian e-commerce platform, managing 300+ databases across 6 regions handling 80K+ requests per second.
- Developed **Go** microservices for transaction processing and inventory management, implementing optimistic locking and distributed transaction patterns that maintained data consistency across PostgreSQL shards serving 100M+ concurrent sellers.
- Designed and implemented **Kubernetes**-based deployment infrastructure using **Helm** charts and **Terraform**, enabling rapid scaling of backend services during flash sales that spiked traffic 10x and processing 500M+ daily transactions.
- Implemented PostgreSQL query optimization across the seller-facing dashboard, rewriting slow queries and adding strategic indexes that reduced backend API response time from 2.8s to under 400ms, improving seller retention by 12%.
- Built real-time monitoring systems using **Prometheus** and **Grafana** to track PostgreSQL replication lag, connection pools, and transaction throughput across all regions, automating alerts that caught production issues before customer impact.
- Established database backup and recovery procedures for multi-region PostgreSQL deployments, implementing automated backups to object storage with point-in-time recovery capabilities ensuring no financial transaction loss even during region failures.
- Collaborated with frontend teams building **React**-based seller dashboards to optimize API contracts, implementing efficient database query patterns and caching strategies that reduced database load by 40% during seasonal peaks.
- Troubleshoot critical production incidents involving PostgreSQL table locks, replication desynchronization, and network partitions across regions, restoring service in under 15 minutes and implementing preventative measures to avoid recurrence.

Altitude Games

Nov 2014–Jul 2018

Full Stack Engineer

- Built backend services for multiplayer games using **Node.js** and **Express**, establishing data persistence patterns with **PostgreSQL** that scaled to support 2M+ concurrent players across global game servers.
- Implemented database optimization strategies including connection pooling, query caching, and schema denormalization that improved in-game transaction latency from 800ms to under 200ms, enhancing player experience across casual and mid-core titles.
- Deployed game infrastructure on **AWS** using **EC2** instances and **RDS** managed PostgreSQL, implementing auto-scaling policies and multi-region failover to maintain 99.5% uptime during peak gaming events and seasonal tournaments.
- Developed observability tooling using custom logging pipelines and basic **Prometheus** metrics collection, enabling the team to identify performance bottlenecks in database queries and game loop processing.
- Established initial backup and disaster recovery procedures for player progression data stored in PostgreSQL, implementing automated daily backups to **S3** that protected against accidental data loss.

EDUCATION

Ateneo de Manila University

Aug 2010–Jul 2014

Bachelor of Science in Computer Science